TAHA KOULAL

Applied AI Engineer · Agent systems & product engineering

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I turn complex workflows into working AI products.

I build across business operations, document intelligence and multi-agent systems — from workflow analysis to backend, interfaces and AI. Grounded in C, Unix and networking.

HOW I WORK

Effective & efficient. Understand the work, design the system, then build and verify. I learn unfamiliar domains quickly and use coding agents with clear permissions and review gates.

TECHNICAL SKILLS

Applied AI & agents
Agent orchestration, MCP, Claude API, RAG, LangChain, ChromaDB, Sentence Transformers, document AI / OCR, scikit-learn, SHAP

Backend & systems
Python, FastAPI, SQLAlchemy, PostgreSQL, REST APIs, JWT, workflow state machines, Supabase / RLS, Docker / Compose, C, C++, Unix / Linux, TCP/IP, WebSockets

Product engineering
TypeScript, JavaScript, React, React Native, Next.js, real-time systems

STRENGTHS

Business process analysis
Workflow digitisation
System design & reliability
Product delivery from zero
RBAC & data boundaries

EDUCATION

42 Network (1337)

Software Engineering
Common Core — in progress

Validated ft_irc, Minishell, Philosophers, ft_printf, Libft and 10+ projects.

LANGUAGES

Arabic — Native
English — Professional
French — Intermediate

EXPERIENCE

PHOVA Technology · Software Engineering Intern

Phova-edge · Business operations & technical AI platform

- Built Phova-edge around how sales, engineering, sourcing and management work: mapped responsibilities and handoffs, then designed and implemented digital workflows.
- Made opportunity progression explicit with a role-gated state machine: **12 statuses, 18 transitions**, notifications, activity history and departmental SLA tracking. Built authentication and user administration.
- Built the design system, most frontend pages and KPI computation across **Next.js, FastAPI and PostgreSQL**, with async SQLAlchemy, Alembic and Docker Compose.
- Converted PDF, Excel and Word documents into structured pump data; normalised French / English technical terminology and located extracted values with OCR for review.
- Connected **RAG retrieval, Random Forest predictions and SHAP explanations** to pump / material recommendations, with hydraulic checks and configurable language / vision providers.

UM6P — IDC Morocco · VR Development Intern

- Developed an immersive virtual visit from prototype to functional delivery using Unity and C#, including interactive 3D environments and navigation.

SELECTED PROJECTS

Factory · Multi-agent AI project factory

Claude Code · MCP · Agent orchestration

- Designed an idea-to-build delegation system: a CEO agent evaluates ideas, specialised CTO modes plan execution, and focused build agents carry it out.
- Made execution traceable through permission gates, milestone reviews, persistent state, audit logs, sandboxing and SLA-based blocker escalation.

Vamo · Football identity mobile product

React Native · Supabase · vamo-l.vercel.app

- Built a football product from zero: persistent player identities and a Football Passport connecting teams, matches, tournaments, attendance and profile history.
- Enforced player-data access in Supabase RLS, including teammate-only phone visibility.

Transcendence · Real-time multiplayer platform

React · TypeScript · WebSockets

- Built authentication, live messaging and synchronised Pong state; implemented frontend state and session behaviour supporting **20+ concurrent games**.

Minishell · Unix shell in C

- Built a working shell in C: quote-aware parsing, expansion, pipelines, redirections, heredocs and builtins, with Unix process control, interactive signals and descriptor cleanup.

Available for full-time and early-stage product opportunities